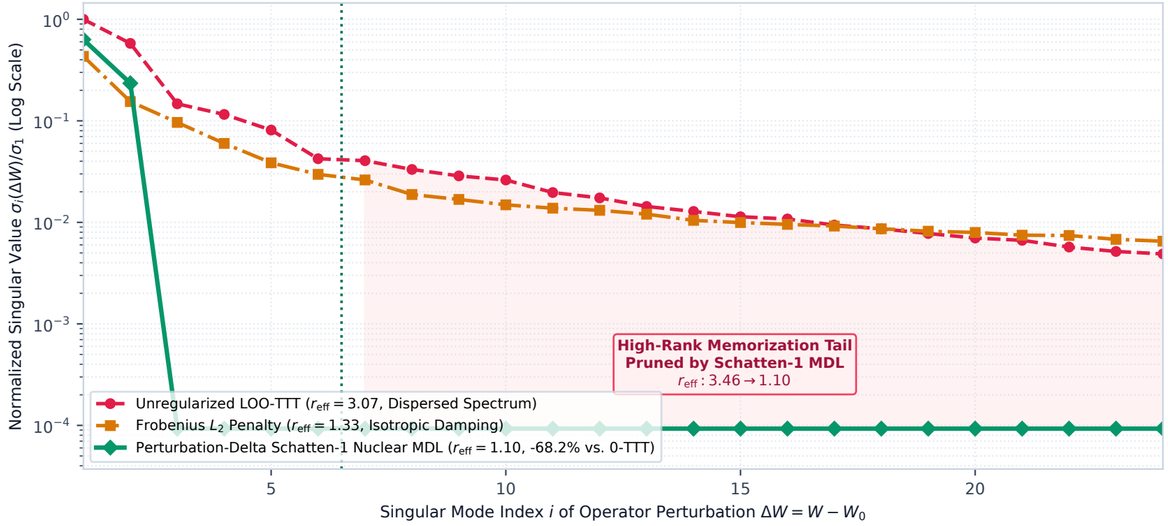
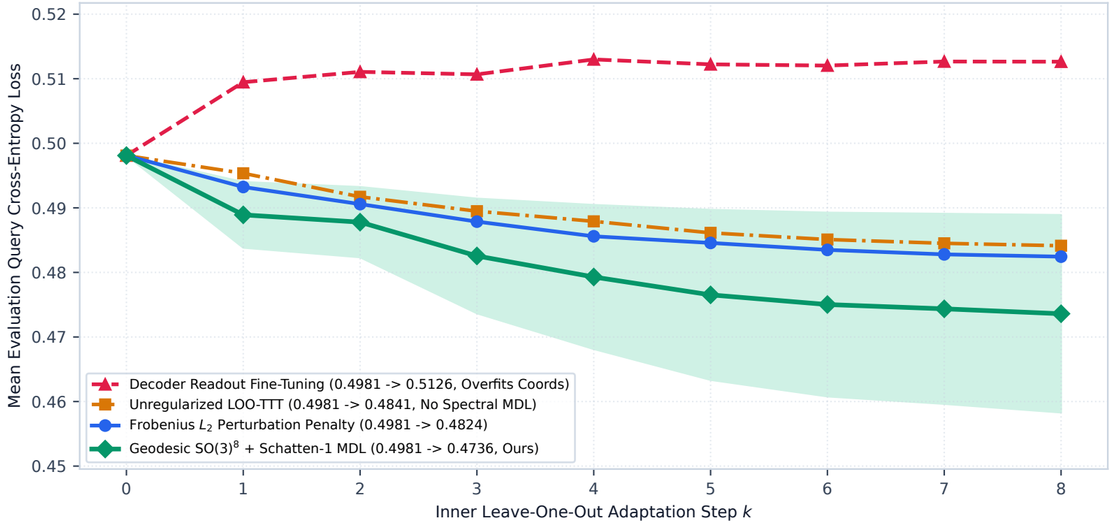
Figure 7 | Perturbation-Delta Schatten-1 Nuclear MDL Spectrum & Geodesic $SO(3)^8$ Test-Time Adaptation

Singular value shrinkage of $\Delta W = W - W_0$ compresses effective rank ($r_{\text{eff}}: 3.46 \to 1.10$) and preserves equivariance under $D_4 \times S_9$ automorphisms

a **Measured Singular Value Spectrum $\sigma_i(\Delta W)$ of Task Perturbation Delta**



b **Live Query Loss Trajectory Across Inner LOO-TTT Steps ($N = 5$ Tasks, ± 1 SEM)**



c **Controlled $D_4 \times S_9$ Group-Equivariance Stress Test on Task 0934a4d8 ($9 \times 3 \to 3 \times 9$)**

Canonical Target (9x3)
Original orientation

7	7	9
7	2	9
7	2	9
7	7	9
4	4	7
4	4	7
6	6	1
6	6	6
1	6	1

Static-Color FT (No S9)
Palette Drift (40.7% Acc)

6	6	6	6	9	9	9	9	3
6	9	9	6	9	9	9	9	9
9	9	9	9	6	6	3	9	3

Geodesic $SO(3)$ + MDL
100.0% Equivariant (27/27)

6	6	6	6	5	5	7	7	3
6	1	1	6	5	5	7	7	7
8	8	8	8	6	6	3	7	3

d **Measured MDL Phase Portrait: Effective Rank r_{eff} vs. Public Evaluation Accuracy**

